

Node.js is a run time environment that allows you to run JS outside the browser, mainly on the server.

Why use node.js?

- Fast
- Scalable
- Handles multiple req efficiently

Q) How node.js is a run time environment on server side. What is VB.

→ A runtime env. is a system that:

- Executes code
- Provides necessary features (like APIs, memory, etc).

Ex-

JS / Browser → run javascript on client side.

Node.js → ————— server ————

Note → Node.js:

- uses V8 engine (from chrome)
- Executes JS outside the browser.

That's why it's called a runtime environment.

→ Node.js is called a runtime env. because it provides the necessary env. to execute JS outside the browser using the V8 engine. It is used on the server side because it enables handling req, databases, and backend logic.

Q) what is VB?

→ VB is a JS engine developed by Google that executes JS code.

Q) Framework - Primarily focus on simplifying the development process by offering a tools, libraries.



## 7 ~~7~~ Main features of Node.js

Asynchronous & Non-Blocking  
Single threaded  
Fast Execution

• Real time capabilities

NPM

Event-Driven Architecture

Cross-platform

Diff. b/w Node.js & Express?

JS



Node.js (Runtime Env)



Express.js (Framework)

Diff. b/w Client Side & Server side.

| Runs on  | User browser   | Server                  |
|----------|----------------|-------------------------|
| Purpose  | UI/Interaction | logic & data processing |
| Speed    | faster (local) | Slower                  |
| Security | less           | More secure             |
| DB       | No             | Yes                     |

Think of a food ordering app:

Client side:

you click buttons, see menu, add items to cart.

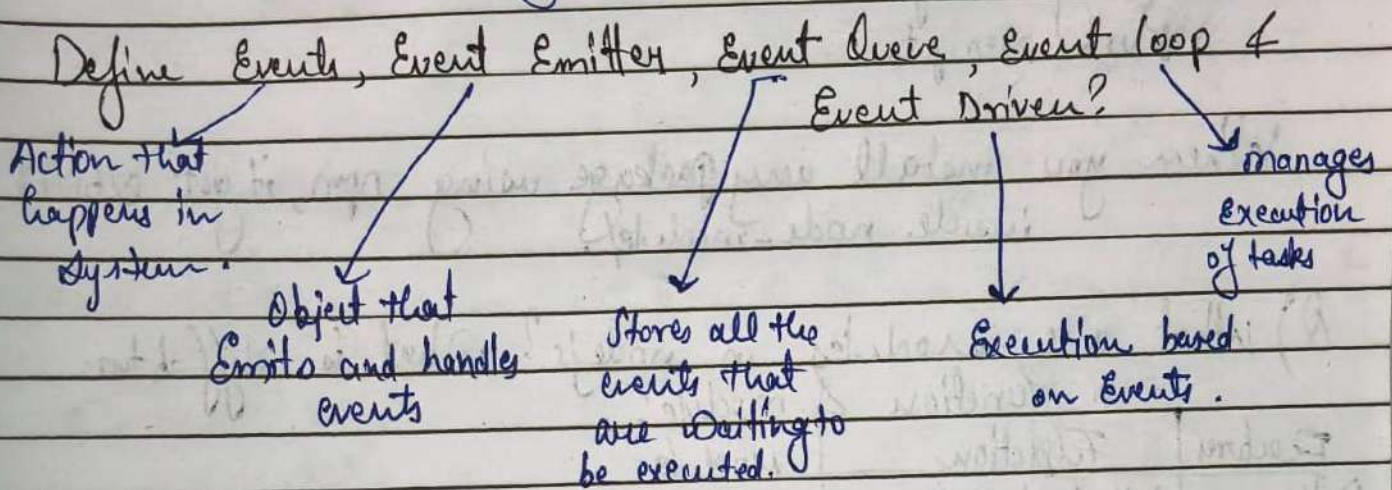
Server side:

processes your order, check availability,  
saves order in database.



Single threaded? - only one task is executed at a time using single thread.

Async? - means a task can start and continue without waiting for other tasks to finish.



### Disadvantages of Node.js!

- Single threaded
- Callback → But ~~not~~ now it can be resolved by promises
- Avoid for applications that involve heavy CPU processing (Images) video processing, Data encryption/Decryption).

### How to set up node.js project?

- Download node.js installer & install it.
- Download VS code
- Create new folder on VS code
- Open terminal → `npm init`
- Create new JS file
- Run your Node.js → `node app.js`



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Q) What is NPM? role of node modules folder?

node package manager is a tool that comes with node.js and is used to install ~~def~~ libraries/packages  
manage dependencies  
Run scripts

When you install any package using npm, it gets stored inside node\_modules.

Q) What are modules in node.js? What is diff. b/w a function & module?

| Features    | Function                     | Module                     |
|-------------|------------------------------|----------------------------|
| Define      | A block of code              | A file containing code     |
| Scope       | Inside a file                | Separate file              |
| Reusability | within same file (or export) | Across multiple files      |
| Size        | Small                        | Can contain many functions |
| Example     | function add()               | math.js                    |

```
function add(a,b){  
  return a+b;  
}
```

```
function add(a,b){  
  return a+b;  
}
```

```
function sub(a,b){  
  return a-b;  
}
```

```
module.export = {add, sub}
```

Q) module is a wrapper function?



## Module wrapper?

When you write this:

```
const x=10;
```

```
console.log(x);
```

// you only see this code.

But internally node runs ↴

```
(function (exports, require, module,  
  -- filename, -- dirname) {
```

```
  const x=10;
```

```
  console.log(x);
```

```
});
```

Why node.js use the module wrapper?

Beoz Node.js want every file to behave like an independent module automatically.

Node.js enforces:

Everything is private.

Only exported things are public.

## Types of modules

① Built in modules — These comes pre-installed with node.js. Ex- fs, http, path, etc.

② user defined module — Module created by you.

```
function add(a,b){  
  return a+b;  
} module.exports = add;
```

```
const add = require('./math');  
console.log(add(2,3));
```

③ Third-party modules — Installed using npm.

What is fs module in Node.js?

It is used to work with files.

It provide methods to read, write, append, update, delete & manage files on the system.



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readFile() / readFileSync

```
fs.readFile('file.txt', 'utf8', (err, data) => {  
  console.log(data);  
});
```

writeFile() / writeFileSync

```
fs.writeFile('file.txt', 'Hello', (err) => {  
  console.log('File written');  
});
```

appendFile()

```
fs.appendFile('file.txt', 'world', (err) => {  
  console.log('Data appended');  
});
```

unlink()

```
fs.unlink('file.txt', (err) => {  
  console.log('File deleted');  
});
```

Rename()

```
fs.rename('file.txt', 'new.txt', (err) => {  
  console.log('File renamed');  
});
```



## Module wrapper?

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// you only see this code.
```

But internally Node runs ↴

```
(function (exports, require, module,  
  -- filename, -- dirname) {  
  console.log(const x=10;  
  console.log(x);  
});
```

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- ① Explain the role of path module? Name some functions of it?
- ② Explain the role of os module? Name some functions of it?
- ③ \_\_\_\_\_ event \_\_\_\_\_ How to handle events in Node?
- ④ What are ~~Event~~ Event Arguments?
- ⑤ diff. b/w. functions & events.
- ⑥ role of http module in node
- ⑦ role of createServer() method of http module?
- ⑧ → createServer() method of the http module is used to create a web server in Node.js.
- ⑨ → is a built-in Node.js module used to create web servers and handle HTTP requests & responses.

| Function                                   | Event                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------|
| Run when called<br>greet()<br>Simple logic | Runs when triggered<br>click, req, emit()<br>Async / event driven system. |

- ④ Arguments are the values passed when an event is emitted, which can be received inside the event listener.

```
event.on('add', (a, b) => {
  console.log(a+b);
});
```

event.emit('add', 8, 3); → args